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|                                                                                   | Document Title: | <b>COMPLETED RESEARCH</b>                      | Revision No.   | <b>00</b>               |
|                                                                                   |                 |                                                | Effective Date | <b>January 26, 2023</b> |
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| <b>(1) Research Program/Project/Study Title:</b> EFFECT OF FERMENTED JANITOR FISH ( <i>Pterygoplichthys pardalis</i> ) ON THE GROWTH PERFORMANCE OF NATIVE CHICKEN. |                                                                                                  |
| <b>Proponent Name:</b>                                                                                                                                              | Rubelle D. Valderrama                                                                            |
| <b>Academic Rank:</b>                                                                                                                                               | <b>Instructor 1</b>                                                                              |
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Program or Project has component (if no proceed to 2)? ☐ Yes ☒ No

|                          |  |
|--------------------------|--|
| <b>Component Title:</b>  |  |
| <b>Component Leader:</b> |  |
| <b>Academic Rank:</b>    |  |

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| <b>(2) Category (please check)</b><br><br><input type="checkbox"/> Social Science/Education /Humanities & other related fields<br><input checked="" type="checkbox"/> Natural/BiologicalSciences<br><input type="checkbox"/> Food Research/<br><input type="checkbox"/> Developmental/ Computer/Engineering/ Architecture<br><input type="checkbox"/> Extension<br><input type="checkbox"/> Poster |  | <b>(3) Research Center</b><br>[Check the concerned center]<br><br><input type="checkbox"/> SSITRDC<br>Social Science and Industrial Technology Research Development Center<br><br><input checked="" type="checkbox"/> LRDC<br>Livestock Research Development Center<br><br><input type="checkbox"/> FFRDC<br>Fisheries and Food Research Development Center<br><br><input type="checkbox"/> CSRDC<br>Crop Science Research Development Center |  |
| <b>(4) Site of implementation</b>                                                                                                                                                                                                                                                                                                                                                                  |  |                                                                                                                                                                                                                                                                                                                                                                                                                                               |  |
| Municipality/City:                                                                                                                                                                                                                                                                                                                                                                                 |  | <b>Dumarao</b>                                                                                                                                                                                                                                                                                                                                                                                                                                |  |
| Province:                                                                                                                                                                                                                                                                                                                                                                                          |  | <b>Capiz</b>                                                                                                                                                                                                                                                                                                                                                                                                                                  |  |
| Region:                                                                                                                                                                                                                                                                                                                                                                                            |  | <b>VI</b>                                                                                                                                                                                                                                                                                                                                                                                                                                     |  |
| <b>(5) Classification</b><br><b>Research:</b> <b>Development:</b><br><br><input checked="" type="checkbox"/> Basic <input type="checkbox"/> Pilot Testing                                                                                                                                                                                                                                          |  | <b>(6) Mode of Implementation</b><br><br><input type="checkbox"/> Single Authorship/Proponent<br><input checked="" type="checkbox"/> Collaborative Research (within campus)<br><input type="checkbox"/> Collaborative Research (intercampus)                                                                                                                                                                                                  |  |

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| <input type="checkbox"/> Applied <input type="checkbox"/> Tech.<br>Promotion/Commercialization                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | <input type="checkbox"/> Collaborative Research (external agency) |
| <b>(7) Sector/Commodity</b><br><br>Agriculture/Animal Science                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | <b>(8) Discipline/Field of Specialization</b>                     |
| <b>(9A) Abstract</b><br><br><p>This study was conducted at Agricultural Project and Research Area (APRA), Capiz State University, Dumarao Campus from March to June 2023. The study aimed to find out the efficacy of the different levels of fish amino acid as water supplement on the growth performance of native chicken in terms of feed consumption, water consumption, gain in weight feed conversion ratio and return above feed and animal cost, to determine whether there is a significant difference among treatment means. There were four treatments, which consisted of the following: Treatment 1: Pure water (control); Treatment 2: 1% fish amino acid and 1L of water; Treatment 3: 2% FAA and 1L of water; Treatment 4: 3% FAA and 1L of water. Results of the study showed that supplementation of the different levels of FAA had a significant effects on native chickens in terms of feed consumption, water consumption, gain in weight, and feed conversion ratio (FCR). However, the return above feed and animal cost revealed that native chicken given 3% FAA yielded the highest return of Php230.06. This suggests that varying levels of FAA have distinct effects on the performance indicators of the birds.</p>                                                                                                                                                                                                                                                                                                                                                                                 |                                                                   |
| <b>(9B) Keywords</b><br>Fish amino acid, supplementation, performance, native chicken, water consumption                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                                                                   |
| <b>(10) Introduction</b><br><p>The Philippines is an agricultural country where most of the Filipinos derive their income from farming specifically poultry farming. But then, resiliency is one of the most important characteristics of native chicken making it a preferred business option. Recently, the Department of Agriculture reported that the prices of poultry and poultry products continued to rise in the market (DA, 2022), and the Philippine population is expected to increase by 115.378 by 2025 (PSA, 2021), therefore increasing demand for chicken and the increasing price of both poultry and poultry products will result to the shortage of chicken produce in the market therefore will consequently result to inflation. Thus, the poultry industry's development must be accelerated. However, the rising cost of production, which is what controls production, disproportionate to the cost of chicken meat. One thing to think about is how to make chicken at the lowest possible cost without sacrificing the product's quality.</p> <p>Globally, organic farming is growing in popularity as a way to reduce the usage of antibiotics and other chemicals used in the raising of livestock and poultry. The misuse of antibiotics might cause a bird's natural defenses to deteriorate. The capacity to combat actual harmful germs and, in most situations, their breeds within the system certain strains of these disease-causing bacteria that are resistant to antibiotics. What's unfortunate is that human health may be harmed by some microorganisms (A.P. Inocencio Farms, 1997).</p> |                                                                   |

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Janitor fish is considered as threat to fresh water biodiversity (D.P. The, 2012) therefore utilizing janitor fish as an amino acid for chicken will not only benefit farmers but will help save biodiversity as well.

Nutritional composition analysis showed that the fish is rich in proteins (16.1%), functional amino acid leucine (15.5%), and macro-minerals phosphorus (2220mg/100g) and potassium (1780mg/100g), the study indicates that the fish is rich in proteins and minerals (Mohanty B.P. et al, 2017). Furthermore, Abad and Siringan found out that janitor fish has a crude protein content of 50.61%, which is comparable to soybean meal, and the study recommended of inclusion of janitor fish in poultry as a protein source.

Finally, absorption of substances through drinking water is faster and higher than food. Therefore, adding materials to water is easier than to food. However, alternative sources of water supplementation in the poultry industry had never been fully recognized, and the poultry raisers relied on the availability of water supplementation in the market, which also recorded a higher cost today.

### **(11) Materials and Methods or Methodology**

The materials used in the study were the following: sixty heads of day old native chicken, feeding and watering troughs, weighing scale, calculator and record book, pale storage container, molasses and rearing pen.

#### **Growth performance assessment on native chicken**

To determine the performance of native chicken, four treatments were utilized in the study and arranged in Completely Randomized Design (CRD) each treatment were replicated three times.

This treatment includes the following:

Treatment 1 - control 100% pure water

Treatment 2 were supplemented with 1% FAA +1 lt of water

Treatment 3 were supplemented with 2% FAA + 1 lt of water

Treatment 4 were supplemented with 3% FAA + 1 lt of water

Sixty (60) 45- day-old native chickens were procured from native chicken raisers at Calinog, Iloilo. Upon arrival of the birds, water with mascovado sugar was given to the birds to reduce transport stress.

Data collected were analyzed using the Analysis of Variance (ANOVA). LSD was used to determine the significant differences between treatment means at 5% level of significance.

### **(12) Results and Discussions**

Table 1.0 presents the Mean Feed Consumption of native chicken supplemented with different levels of FAA. Data shows that the birds fed with 3% FAA have the highest feed consumption with a mean of 4594.60g. This was followed by the birds fed with 2% FAA and 1% FAA with a mean of 4374.13g and 4339.93g respectively. Birds given with pure water attained the lowest feed consumption with a mean of 4286.60g.

Analysis of variance revealed that there was a significant difference among treatment means at 5% level of significance. LSD of the study reveals that quails fed with 3% FAA showed the highest results in feed consumption.

Table 1.0. Mean Feed Consumption (g) of native chicken supplemented with different levels of amino acid

| TREATMENTS                           | REPLICATION |         |         | TOTAL    | MEAN                 |
|--------------------------------------|-------------|---------|---------|----------|----------------------|
|                                      | R1          | R2      | R3      |          |                      |
| T1-100% water                        | 4343.40     | 4349.00 | 4167.40 | 12859.80 | 4286.60 <sup>b</sup> |
| T2-1% (10ml)<br>FAA + 1L of<br>water | 4327.40     | 4340.40 | 4352.00 | 13019.80 | 4339.93 <sup>b</sup> |
| T3-2% (20ml)<br>FAA + 1L of<br>water | 4364.60     | 4375.80 | 4382.00 | 13122.40 | 4374.13 <sup>b</sup> |
| T4-3% (30ml)<br>FAA + 1L of<br>water | 4571.40     | 4606.40 | 4606.00 | 13783.80 | 4594.60 <sup>a</sup> |
| Grand Total                          |             |         |         | 52785.80 |                      |
| Grand Mean                           |             |         |         |          | 4398.82              |

Table 2.0 presents the water Consumption of native chicken supplemented with different levels of FAA. Data shows that the birds fed with 3% FAA have the highest water consumption with a mean of 5227.60ml. This was followed by the birds fed with 2% and 1% FAA with a mean of 5057.87ml and 5007.53ml respectively. Birds given with pure water attained the lowest egg weight with a mean of 4906.73ml.

Analysis of variance of the study revealed that there was a significant difference among treatment means. This means that the different levels of FAA supplementation affects the water consumption of the experimental birds.

Table 2.0. Mean Water Consumption (ml) of native chicken supplemented with different levels of amino acid

| TREATMENTS                           | REPLICATION |         |         | TOTAL    | MEAN                  |
|--------------------------------------|-------------|---------|---------|----------|-----------------------|
|                                      | R1          | R2      | R3      |          |                       |
| T1-100% water                        | 5021.00     | 4430.80 | 4868.40 | 14720.20 | 4906.73 <sup>b</sup>  |
| T2-1% (10ml)<br>FAA + 1L of<br>water | 4976.80     | 5019.00 | 5035.80 | 15022.60 | 5007.53 <sup>b</sup>  |
| T3-2% (20ml)<br>FAA + 1L of<br>water | 5019.40     | 5256.80 | 4897.40 | 15173.60 | 5057.87 <sup>ab</sup> |
| T4-3% (30ml)<br>FAA + 1L of<br>water | 5259.00     | 5172.80 | 5251.00 | 15682.80 | 5227.60 <sup>a</sup>  |
| Grand Total                          |             |         |         | 60599.20 |                       |
| Grand Mean                           |             |         |         |          | 5049.93               |

Table 3.0 presents the gain in weight of native chicken supplemented with different levels of FAA. Data shows that the birds given with 3% level of FAA have the highest weight gained with a mean of 1134.51g. This was followed by the birds given with 2% and 1% level FAA with a mean of

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1052.49g and 889.04g respectively. Birds given with pure water attained the lowest weight gain with a mean of 813.35g.

Analysis of variance of the study revealed that there was a significant difference among treatment means. This means that the different levels of FAA supplementation significantly affect weight gained of the experimental birds.

Table 3.0. Gain in Weight(g) of native chicken supplemented with different levels of amino acid

| TREATMENTS                           | REPLICATION |         |         | TOTAL    | MEAN                 |
|--------------------------------------|-------------|---------|---------|----------|----------------------|
|                                      | R1          | R2      | R3      |          |                      |
| T1-100% water                        | 809.88      | 872.24  | 757.92  | 2440.04  | 813.35 <sup>b</sup>  |
| T2-1% (10ml)<br>FAA + 1L of<br>water | 932.64      | 943.24  | 791.24  | 2667.12  | 889.04 <sup>b</sup>  |
| T3-2% (20ml)<br>FAA + 1L of<br>water | 967.00      | 1041.96 | 1148.52 | 3157.48  | 1052.49 <sup>a</sup> |
| T4-3% (30ml)<br>FAA + 1L of<br>water | 1122.76     | 1164.96 | 1115.80 | 3403.52  | 1134.51 <sup>a</sup> |
| Grand Total                          |             |         |         | 11668.16 |                      |
| Grand Mean                           |             |         |         |          | 972.35               |

Table 4.0 presents the Feed Conversion Ratio of native chicken supplemented with different levels of FAA supplementation. Data shows that the birds given 3% FAA supplement have the lowest FCR with a mean of 4.05, followed by the birds given with 2% and 1% FAA supplement with the mean of 4.18 and 4.91 respectively. Birds given pure water have the highest FCR with the mean of 5.28.

Analysis of variance revealed that there was a significant difference among treatment means. This means that the different levels of FAA supplementation significantly affect the feed conversion ratio of the experimental birds.

Table 4.0. Mean Feed Conversion Ratio of native chicken supplemented with different levels of amino acid

| TREATMENTS                           | REPLICATION |      |      | TOTAL | MEAN              |
|--------------------------------------|-------------|------|------|-------|-------------------|
|                                      | R1          | R2   | R3   |       |                   |
| T1-100% water                        | 5.36        | 4.99 | 5.50 | 15.85 | 5.28 <sup>b</sup> |
| T2-1% (10ml)<br>FAA + 1L of<br>water | 4.64        | 4.60 | 5.50 | 14.74 | 4.91 <sup>b</sup> |
| T3-2% (20ml)<br>FAA + 1L of<br>water | 4.51        | 4.20 | 3.82 | 12.53 | 4.18 <sup>a</sup> |
| T4-3% (30ml)<br>FAA + 1L of<br>water | 4.07        | 3.95 | 4.13 | 12.15 | 4.05 <sup>a</sup> |
| Grand Total                          |             |      |      | 55.27 |                   |
| Grand Mean                           |             |      |      |       | 4.61              |

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Table 5.0. Presents the return above feed and animal cost of native chicken supplemented with different levels of FAA supplementation. Data shows that birds given with 3% level FAA supplement had the highest net income of Php 230.06, followed by the birds given 2% and 1% level of supplement with the net income of Php 203.94 and Php147.48 respectively. Lowest net income was noted on birds given pure water with a net income of Php 130.98.

**Table 16. Return Above Feed and Animal Cost of native chickens fed with Different levels of FAA.**

| PARTICULARS             | TREATMENTS    |               |               |               |
|-------------------------|---------------|---------------|---------------|---------------|
|                         | T1            | T2            | T3            | T4            |
| <b>SALES OF BIRDS</b>   | <b>321</b>    | <b>346.49</b> | <b>409.46</b> | <b>443.21</b> |
| <b>PRODUCTION COSTS</b> |               |               |               |               |
| <b>BIRDS</b>            | <b>70</b>     | <b>70</b>     | <b>70</b>     | <b>70</b>     |
| <b>FEEDS</b>            | <b>120.02</b> | <b>121.51</b> | <b>122.48</b> | <b>128.65</b> |
| <b>JANITOR FISH</b>     | <b>-</b>      | <b>0.42</b>   | <b>1.69</b>   | <b>3.92</b>   |
| <b>MOLASSES</b>         | <b>-</b>      | <b>0.42</b>   | <b>1.69</b>   | <b>3.92</b>   |
| <b>SUBTOTAL</b>         | <b>190.02</b> | <b>199.01</b> | <b>202.52</b> | <b>213.15</b> |
| <b>NET INCOME</b>       | <b>130.98</b> | <b>147.48</b> | <b>203.94</b> | <b>230.06</b> |

### (13) Conclusions and Recommendations

Based on the results of the study, the following conclusions were made:

1. Supplementation of FAA at 3% level had significantly affect the Feed Conversion Ration, Feed Consumption and Gain in Weight of birds .
2. Water consumption of birds was found to be significant with the highest mean water consumption observed in the group supplemented with 3% FAA.

#### Recommendations

Based on the result of the study, the following recommendations are made:

1. Janitor fish's future source sustainability could be a great issue to insure the sustenance of FAA production. Therefore it is recommended to culture Janitor fish before the start of the future researches that would involve mainly the aforementioned fish.
2. Maybe conduct a study with a broader scope

### (14) Expected Output

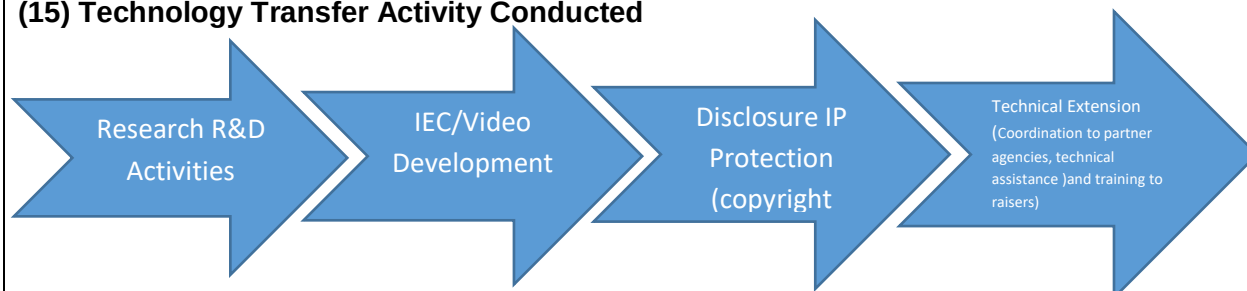
- IEC and product development

|             |                                                                             |
|-------------|-----------------------------------------------------------------------------|
| Product     | Native chicken                                                              |
| Publication | IEC materials                                                               |
| Patent      | Utility models for Janitor Fish Amino Acid for the growth of native chicken |

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| People and Services | Helping farmers with their native farming business                                                                |
| Places and Linkages | Partnership with the poultry farmers, and LGU.<br>Linkages with the DTI and LGU for the marketing of the product. |

#### (15) Technology Transfer Activity Conducted



#### (16) Economic Analysis

#### (17) Acknowledgments

The researchers would like to acknowledge the Research Development and Extension Office of Capiz State University, Dumarao Satellite College for allowing the conduct of this study.

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